

Delivery 2: F3 - Hyperbolic Sine $\sinh(x)$

From-scratch Python implementation with Tkinter GUI

Arvind Lakshmanan (40310757)

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Assigned function: **F3:** $\sinh(x)$

D2 Roadmap

- ➊ Modify D1 implementation into a from-scratch version.
- ➋ Add a Tkinter graphical user interface.
- ➌ Handle exceptions with helpful messages.
- ➍ Prepare public version-control evidence.
- ➎ Update requirements based on the D2 implementation.

D1 to D2 Change

D1 implementation

- Textual interface.
- Used direct formula:

$$\sinh(x) = \frac{e^x - e^{-x}}{2}$$

- Used `math.exp`.

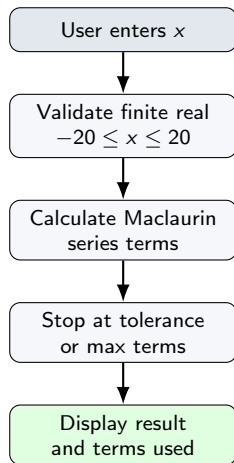
D2 implementation

- Tkinter GUI.
- Uses Maclaurin series:

$$\sinh(x) = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$$

- No Python math-library function for calculation.

Problem 5: From-scratch Algorithm



The implementation separates validation, calculation, stopping logic, and output.

Problem 5: Series Recurrence

```
term_0 = x
sum = x

term_n = term_(n-1) * x^2 / ((2n) * (2n + 1))
sum = sum + term_n

stop when:
|term_n| <= tolerance * (1 + |sum|)
```

Scratch implementation evidence

The calculation avoids `math.sinh`, `math.exp`, `math.factorial`, NumPy, and SciPy.

Problem 5: GUI and Exceptions

Tkinter GUI includes

- input box for x ,
- Calculate button,
- Clear button,
- Exit button,
- result and status area.

Handled errors

- blank input,
- non-numeric input,
- NaN or infinity,
- out-of-range input,
- convergence failure.

Problem 5: Manual Validation Evidence

Input	Approximate result	Case
0	0	zero
1	1.1752011936438	positive
-1	-1.1752011936438	negative
3	10.0178749274099	larger input
20	242582597.704895	upper boundary

Problem 6: Version Control Evidence

Hosting service

Use a public GitHub repository and replace the placeholder before submission:

```
https://github.com/<your-username>/soen6011-d2-f3-sinh
```

High-quality commit examples

- docs: add D2 README with setup and run instructions
- feat: implement from-scratch Maclaurin series for sinh
- feat: add Tkinter GUI with input validation and helpful errors
- docs: update D2 requirements and traceability

Problem 7: Updated Requirements

ID	Requirement	Priority
REQ-F3-D2-01	The system shall provide a Tkinter graphical user interface.	High
REQ-F3-D2-03	The system shall support inputs in $[-20, 20]$.	High
REQ-F3-D2-04	The system shall calculate $\sinh(x)$ using a from-scratch Maclaurin series.	High
REQ-F3-D2-05	The system shall not use prohibited math-library functions for calculation.	High
REQ-F3-D2-07	The system shall provide helpful errors for invalid inputs.	High
REQ-F3-D2-12	The source code shall be placed in a public distributed version control system with a README.	High

Problem	Prompt type	How output was used
P5	Implementation-generation prompt with constraints	Helped structure scratch algorithm, GUI, and exceptions.
P6	Documentation and version-control planning prompt	Helped prepare README and commit-message examples.
P7	Requirements-revision prompt	Helped update requirements and trace D1 changes to D2.

For D2, the F3 calculator was modified from a D1 textual implementation into a Tkinter GUI application. The calculation now uses a from-scratch Maclaurin series, includes exception handling with helpful messages, has repository documentation, and includes an updated list of requirements aligned with the D2 constraints.